

Tejas Rao | Applied Dynamical Systems – Homework 5

Question 1

Homework 5: Q1) Given $\dot{x} = Ax$, where $\dot{x} = \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix}$
 A is a constant 3×3 matrix, with eigenvalues $\lambda_1, \lambda_2, \lambda_3$
& $x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

a) For $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ st $\lambda_1 \neq \lambda_2 \neq \lambda_3$:

Now, A is diagonalisable, and the canonical form Π is

$$\Rightarrow \Pi = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

b) Given, $\lambda_{1,2} = \alpha + i\beta \in \mathbb{C}$, $\lambda_3 \in \mathbb{R}$. Here,

$$\Pi = \begin{pmatrix} \alpha & \beta & 0 \\ -\beta & \alpha & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

c) Given $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$, $\lambda_1 = \lambda_2 = \lambda$

1) If A is diagonalisable:

$$\Pi = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

2) If A isn't diagonalisable:

$$\Pi = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

d) $\lambda_1 = \lambda_2 = \lambda_3 = \lambda$, $\lambda \in \mathbb{R}$, there are 3 possibilities:

1) When 3 eigenvectors are independent. $\Pi = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$

2) 2 independent eigenvectors: $\pi = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$

3) When there is only 1 independent eigenvectors:

$$\pi = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$$

Question 2

Q2) Given $A = \begin{pmatrix} c & 0 & a \\ 0 & b & 0 \\ -a & 0 & c \end{pmatrix}$

a) To find eigenvalues: $\det(A - \lambda I_3) = 0$

$$\Rightarrow \begin{vmatrix} c-\lambda & 0 & a \\ 0 & b-\lambda & 0 \\ -a & 0 & -\lambda \end{vmatrix} = (b-\lambda) \left[(c-\lambda)(c-\lambda) + a^2 \right] = 0$$

$$= (b-\lambda) \left[\lambda^2 - c\lambda + a^2 \right] = 0$$

$$\Rightarrow \lambda_1 = b, \quad \lambda_{2,3} = \frac{c \pm \sqrt{c^2 - 4a^2}}{2}, \quad \lambda_3 = \frac{c - \sqrt{c^2 - 4a^2}}{2}$$

b) To find constraints a, b, c that fulfill the forms in Q1

Now, the discriminant $D = c^2 - 4a^2$

For Q1a), $D > 0 \Rightarrow c^2 > 4a^2$, (Real eigenvalues)

$$\text{Also, } b \neq \lambda_3 \Rightarrow b \neq \frac{c \pm \sqrt{c^2 - 4a^2}}{2} \Rightarrow b^2 - bc + a^2 \neq 0$$

For Q1b), $D < 0 \Rightarrow c^2 < 4a^2$ (No other constraints, $\lambda_{1,2} \in \mathbb{C}, \lambda_3 \in \mathbb{R}$)

For Q1c) $D=0 \Rightarrow c^2=4a^2$ (Repeated eigenvalues)

For Q1d) $b=\lambda_2=\lambda_3 \Rightarrow b^2-cb+a^2=0$

c) A F.P. is non hyperbolic if the eigenvalues has no real part.

1) If $\lambda_1=0: b=0$

2) If $\lambda_{2,3}$ have zero real part.

• If $\lambda_{2,3} \in \mathbb{R}$, one is 0 $\Rightarrow 0^2 \cdot c \cdot 0 + c^2 = 0 \Rightarrow a=0$

• If $\lambda_{2,3} \in \mathbb{C}$, $c^2 < 4a^2$ the real part is $\frac{c}{2} \Rightarrow c=0$

Thus, the f.p is non-hyperbolic when $a=0$ or $b=0$ or $c=0$

Question 3

Part a)

Q3) Given the linear Autonomous system $\dot{\vec{x}} = A\vec{x}$, the general solⁿ is: $\vec{x}(t) = c_1 e^{\lambda_1 t} \vec{v}_1 + c_2 e^{\lambda_2 t} \vec{v}_2 + c_3 e^{\lambda_3 t} \vec{v}_3$, where $\vec{v}_i$ is the corresponding Eigenvector for eigenvalue λ_i , $i \in \{1, 2, 3\}$. To verify:

$$\text{Differentiate w.r.t time } t: \frac{d}{dt}(\vec{x}(t)) = \frac{d}{dt}(c_1 e^{\lambda_1 t} \vec{v}_1 + c_2 e^{\lambda_2 t} \vec{v}_2 + c_3 e^{\lambda_3 t} \vec{v}_3)$$

$$\Rightarrow \dot{\vec{x}} = c_1 \lambda_1 e^{\lambda_1 t} \vec{v}_1 + c_2 \lambda_2 e^{\lambda_2 t} \vec{v}_2 + c_3 \lambda_3 e^{\lambda_3 t} \vec{v}_3 = \sum_{i=1}^3 c_i \lambda_i e^{\lambda_i t} \vec{v}_i = L.H.S$$

$$\text{For } A\vec{x} = R.H.S: A\vec{x} = A\left(\sum_{i=1}^3 c_i e^{\lambda_i t} \vec{v}_i\right) = \left(\sum_{i=1}^3 c_i e^{\lambda_i t} (A\vec{v}_i)\right).$$

However, by defⁿ of eigenvectors, $A\vec{v}_i = \lambda_i \vec{v}_i$

$$\Rightarrow A\vec{x} = \left(\sum_{i=1}^3 c_i e^{\lambda_i t} \lambda_i \vec{v}_i\right) = R.H.S \text{ (from above)}$$

Thus, as $\dot{\vec{x}} = A\vec{x}$, $\vec{x}(t) = \sum_{i=1}^3 c_i e^{\lambda_i t} (\vec{v}_i)$ satisfies the governing differential equation.

Part b)

b) For a sink: $\lambda_1, \lambda_2, \lambda_3 < 0$ (All eigenvalues are -ve)

For a source: $\lambda_1, \lambda_2, \lambda_3 > 0 \Rightarrow$ All eigenvalues are +ve

For a saddle: There must exist one eigenvalue such that its sign is different from the signs of the other two.

→ Case 1: 2 positive eigenvalues, 1 negative

→ Case 2: 2 negative eigenvalues, 1 positive

Part c)

For a Sink:

```
In[ ]:= (* 3D Sink: all trajectories flow into the origin *)

ClearAll["Global`*"];
A = DiagonalMatrix[{-1, -2, -3}];

(* streamlines *)
pStream = StreamPlot3D[
  A . {x, y, z},
  {x, -2, 2}, {y, -2, 2}, {z, -2, 2},
  StreamPoints -> Fine,
  StreamColorFunction -> None,
  StreamStyle -> Directive[VBlue, Medium],
  BoxRatios -> {2, 1, 1},
  PlotRange -> All
];

(* Arrows: ChagPt used to refine it now*)

pts = Flatten[
  Table[{x0, y0, z0}, {x0, -1, 1, 0.5}, {y0, -1, 1, 0.5}, {z0, -1, 1, 0.5}],
  2
];

scale = 0.15; (* arrow length multiplier *)

pArrows = Graphics3D[
  {
    Directive[Red, Opacity[0.6], Thick],
    Arrowheads[0.03],
    Arrow /@ Table[
      {
        p,
        p + scale (A . p)
      },
    ],
  },
]
```



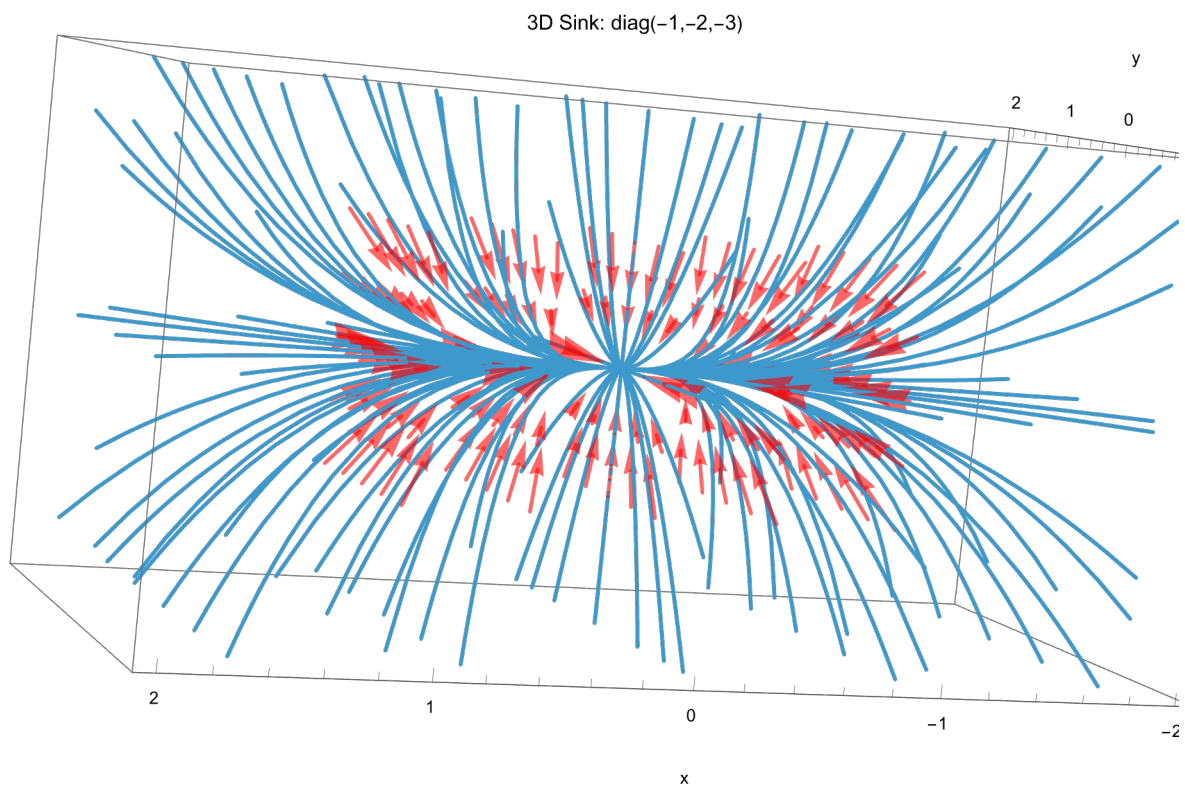
```

    {p, pts}
  ]
},
PlotRange → {{-2, 2}, {-2, 2}, {-2, 2}},
BoxRatios → {2, 1, 1}
];

Show[
  pStream, pArrows,
  AxesLabel → {"x", "y", "z"},
  PlotLabel → "3D Sink: diag(-1,-2,-3)"
]

```

Out[]=



For a Source:

```

(* SOURCE (all eigenvalues positive)
   Canonical form:  $x' = A x$  with  $A = \text{DiagonalMatrix}[\{1,2,3\}]$  *)

ClearAll["Global`*"];

A = DiagonalMatrix[{1, 2, 3}];

(* 1) Streamlines *)
pStream = StreamPlot3D[
  A . {x, y, z},
  {x, -2, 2}, {y, -2, 2}, {z, -2, 2},
  StreamPoints → Fine,
  StreamColorFunction → None,
  StreamStyle → Directive[Vblue, Medium],
  BoxRatios → {2, 1, 1},
  PlotRange → All
];

(* 2) True arrows (central subset) *)
pts = Flatten[
  Table[{x0, y0, z0}, {x0, -1, 1, 0.5}, {y0, -1, 1, 0.5}, {z0, -1, 1, 0.5}],
  2
];

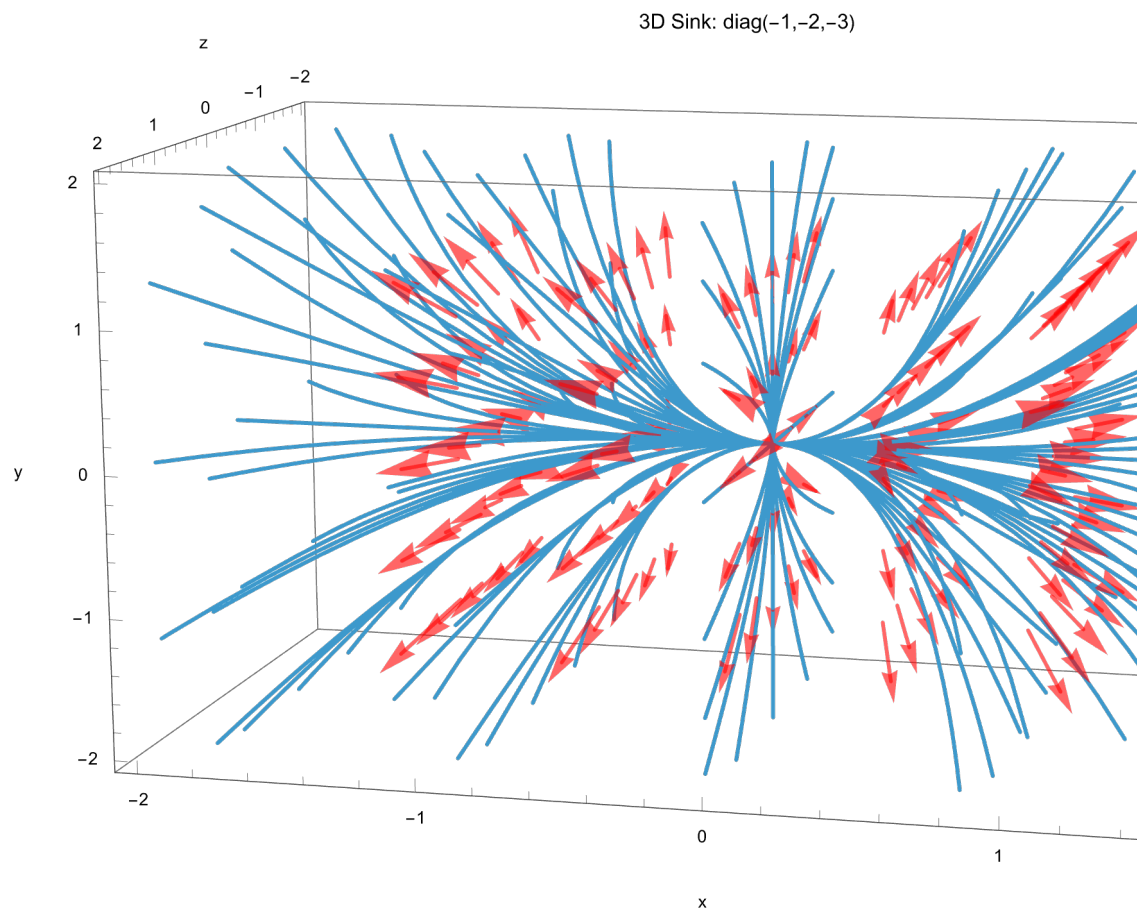
scale = 0.2;

pArrows = Graphics3D[
  {
    Directive[Red, Opacity[0.6], Thick],
    Arrowheads[0.03],
    Arrow /@ Table[{p, p + scale (A . p)}, {p, pts}]
  },
  PlotRange → {{-2, 2}, {-2, 2}, {-2, 2}},
  BoxRatios → {2, 1, 1}
];

Show[
  pStream, pArrows,
  AxesLabel → {"x", "y", "z"},
  PlotLabel → "3D Sink: diag(-1,-2,-3)"
]

```

Out[]=



For a Saddle, there are 2 cases

```

In[ ]:= (*Saddle, 2 eigenvalues positive, one negative*)
ClearAll["Global`*"];

A = DiagonalMatrix[{1, 2, -3}];

(* 1) Streamlines *)
pStream = StreamPlot3D[
  A . {x, y, z},
  {x, -2, 2}, {y, -2, 2}, {z, -2, 2},
  StreamPoints → Fine,
  StreamColorFunction → None,
  StreamStyle → Directive[Vblue, Medium],
  BoxRatios → {2, 1, 1},
  PlotRange → All
];

(* 2) True arrows (central subset) *)
pts = Flatten[
  Table[{x0, y0, z0}, {x0, -1, 1, 0.5}, {y0, -1, 1, 0.5}, {z0, -1, 1, 0.5}],
  2
];

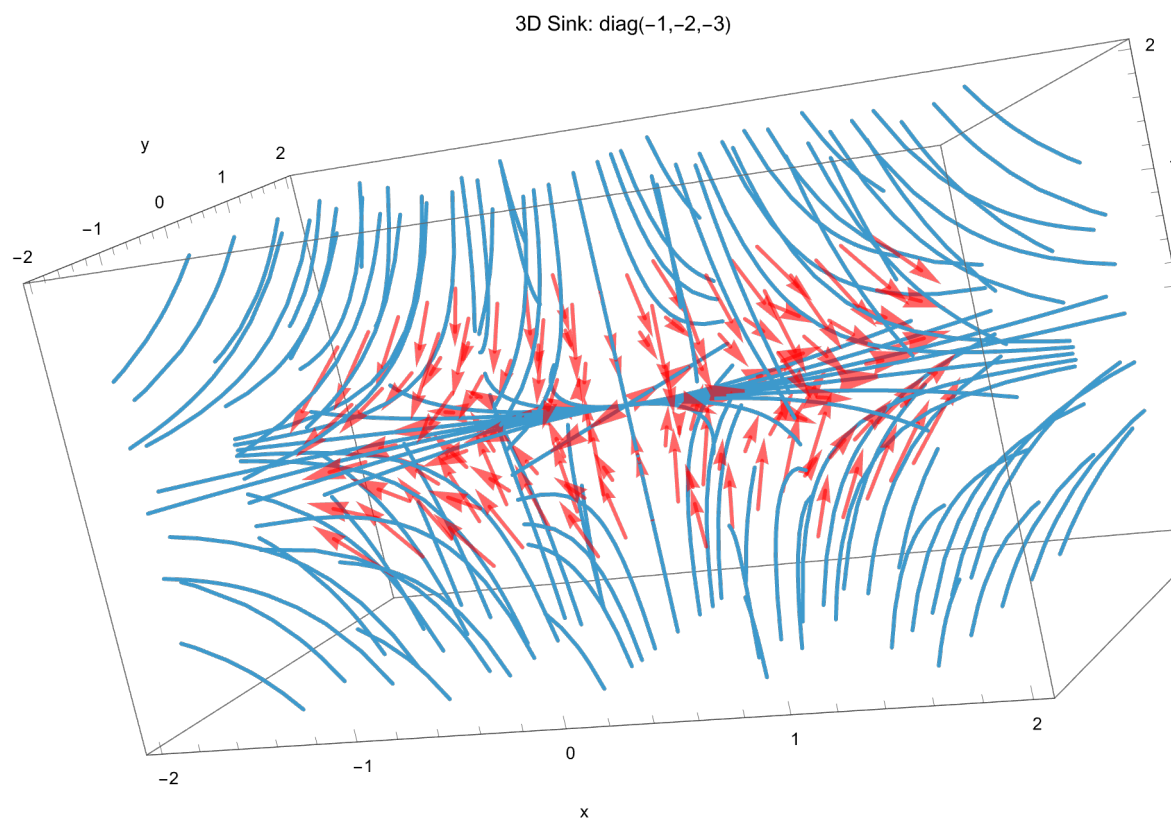
scale = 0.2;

pArrows = Graphics3D[
  {
    Directive[Red, Opacity[0.6], Thick],
    Arrowheads[0.03],
    Arrow /@ Table[{p, p + scale (A . p)}, {p, pts}]
  },
  PlotRange → {{-2, 2}, {-2, 2}, {-2, 2}},
  BoxRatios → {2, 1, 1}
];

Show[
  pStream, pArrows,
  AxesLabel → {"x", "y", "z"},
  PlotLabel → "3D Sink: diag(-1,-2,-3)"
]

```

Out[]=



For the other case

```

In[ ]:= (*Saddle, 2 eigenvalues positive, one negative*)
ClearAll["Global`*"];

A = DiagonalMatrix[{-1, -2, 3}];

(* 1) Streamlines *)
pStream = StreamPlot3D[
  A . {x, y, z},
  {x, -2, 2}, {y, -2, 2}, {z, -2, 2},
  StreamPoints → Fine,
  StreamColorFunction → None,
  StreamStyle → Directive[Vblue, Medium],
  BoxRatios → {2, 1, 1},
  PlotRange → All
];

(* 2) True arrows (central subset) *)
pts = Flatten[
  Table[{x0, y0, z0}, {x0, -1, 1, 0.5}, {y0, -1, 1, 0.5}, {z0, -1, 1, 0.5}],
  2
];

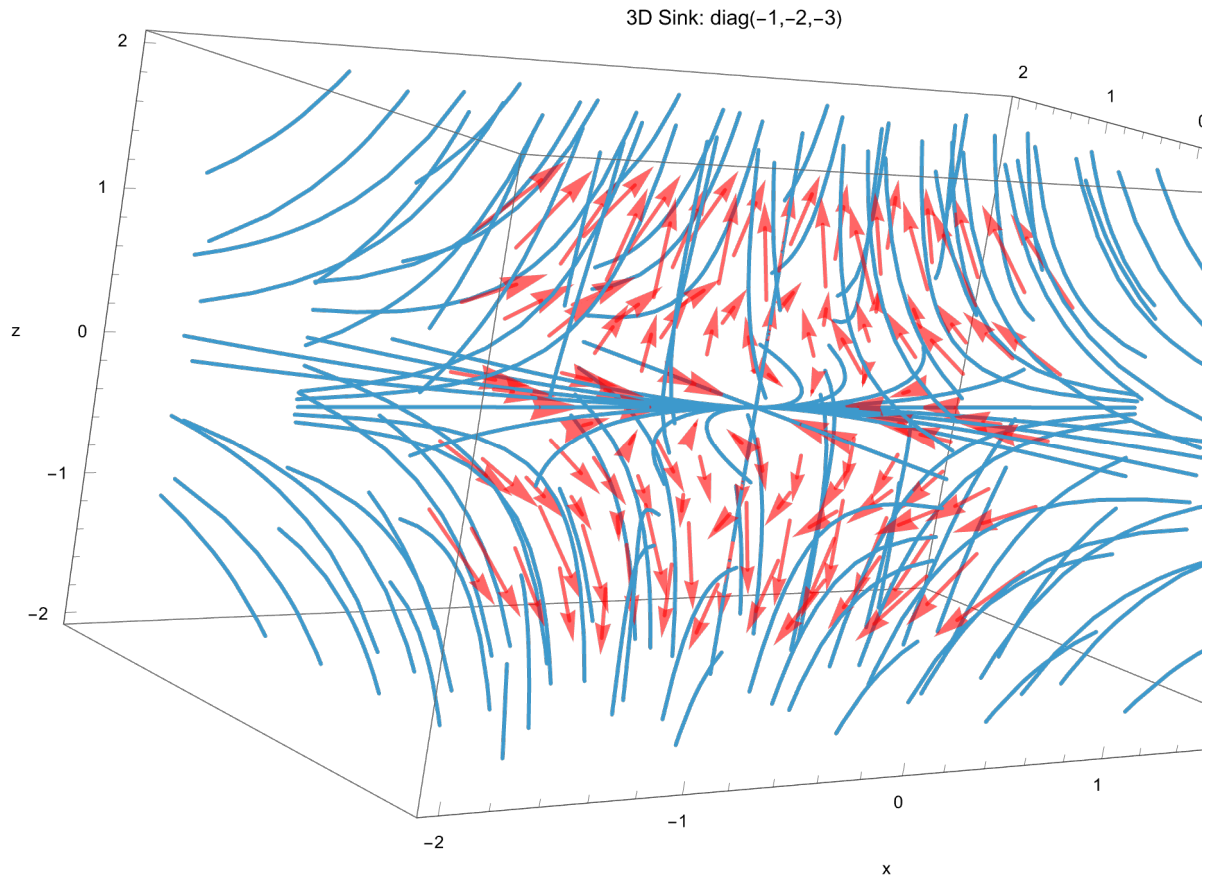
scale = 0.2;

pArrows = Graphics3D[
  {
    Directive[Red, Opacity[0.6], Thick],
    Arrowheads[0.03],
    Arrow /@ Table[{p, p + scale (A . p)}, {p, pts}]
  },
  PlotRange → {{-2, 2}, {-2, 2}, {-2, 2}},
  BoxRatios → {2, 1, 1}
];

Show[
  pStream, pArrows,
  AxesLabel → {"x", "y", "z"},
  PlotLabel → "3D Sink: diag(-1,-2,-3)"
]

```

Out[]=



Question 4

Part a)

Q4) Call system 1 S_1 s.t. $S_1: \begin{pmatrix} \dot{x}_1 \\ \dot{y}_1 \end{pmatrix} = \begin{pmatrix} y_1 \\ -\omega_1^2 x_1 \end{pmatrix}$ and system 2 s.t. $\begin{pmatrix} \dot{x}_2 \\ \dot{y}_2 \end{pmatrix} = \begin{pmatrix} y_2 \\ -\omega_2^2 x_2 \end{pmatrix}$

From above, we can write the following Eqⁿs: $\dot{x}_1 = 0x_1 + 1y_1 + 0x_2 + 0y_2$
 $\dot{y}_1 = -\omega_1^2 x_1 + 0y_1 + 0x_2 + 0y_2$, $\dot{x}_2 = 0x_1 + 0y_1 + 0x_2 + 1y_2$, $\dot{y}_2 = 0x_1 + 0y_1 - \omega_2^2 x_2 + 0y_2$

we can thus rewrite the 2 decoupled systems S_1 & S_2 as a system S

With state vector $x^* = [x_1, y_1, x_2, y_2]^T$ s.t. $\dot{S} = A x^*$, where A is a 4×4 matrix given by $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\omega_1^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\omega_2^2 & 0 \end{bmatrix}$

Part b)

b) For eigenvalues of S_1 : $\dot{x} = \begin{bmatrix} 0 & 1 \\ -\omega_1^2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$, solve $\begin{vmatrix} 0-\lambda & 1 \\ -\omega_1^2 & 0-\lambda \end{vmatrix} = 0$

$$\Rightarrow \lambda^2 + \omega_1^2 = 0 \Rightarrow \lambda_{1,2} = \pm i\omega_1$$

Here, $(x_1, y_1) = (0, 0)$ is a F.P. As eigenvalues are purely complex, according to the Tr-Det plane classification, this F.P. $(x_1, y_1) = (0, 0)$ is a centre.

Conducting a similar analysis for S_2 : $\dot{x} = \begin{bmatrix} 0 & 1 \\ -\omega_2^2 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$

For eigenvalues, $\begin{vmatrix} -\lambda & 1 \\ -\omega_2^2 & -\lambda \end{vmatrix} \Rightarrow \lambda_{3,4} = \pm i\omega_2 \Rightarrow \lambda_3, \lambda_4 \in \mathbb{C}$

Also, F.P for S_2 is $(x_2, y_2) = (0, 0)$. Given that the eigenvalues are purely imaginary, By Tr-Det plane classification, this F.P is a centre.

Part c)

c) Given, $\begin{pmatrix} x_1(t) \\ y_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = \begin{pmatrix} a_1 \cos \omega_1 t + b_1 \sin \omega_1 t \\ \omega_1 (-a_1 \sin \omega_1 t + b_1 \cos \omega_1 t) \\ a_2 \cos \omega_2 t + b_2 \sin \omega_2 t \\ \omega_2 (-a_2 \sin \omega_2 t + b_2 \cos \omega_2 t) \end{pmatrix}$ To verify this, check if it satisfies eq^{ns} for S_1 & S_2

For S_1 : Check $x_1(t)$

Now, $\frac{d}{dt} x_1(t) = \frac{d}{dt} (a_1 \cos \omega_1 t + b_1 \sin \omega_1 t) = -a_1 \omega_1 \sin \omega_1 t + b_1 \omega_1 \cos \omega_1 t$

$\Rightarrow \frac{d}{dt} x_1(t) = \omega_1 [-a_1 \sin \omega_1 t + b_1 \cos \omega_1 t] = y_1(t) \Rightarrow \therefore \dot{x}_1 = y_1(t)$

Check $y_1(t)$: $\frac{d}{dt} y_1(t) = \frac{d}{dt} [\omega_1 (-a_1 \sin \omega_1 t + b_1 \cos \omega_1 t)]$

$\frac{d}{dt} y_1(t) = \omega_1 [-a_1 \omega_1 \cos \omega_1 t - b_1 \sin \omega_1 t] = -\omega_1^2 [a_1 \cos \omega_1 t + b_1 \sin \omega_1 t]$

$\Rightarrow \frac{d}{dt} y_1(t) = -\omega_1^2 x_1 \Rightarrow \text{Verified for } S_1$

For S_1 : Check $x_2(t)$

$$\text{Now } \frac{d}{dt} x_2(t) = \frac{d}{dt} (a_2 \cos \omega_2 t + b_2 \sin \omega_2 t) = -a_2 \omega_2 \sin \omega_2 t + b_2 \omega_2 \cos \omega_2 t$$

$$\Rightarrow \frac{d}{dt} x_2(t) = \omega_2 [-a_2 \sin \omega_2 t + b_2 \cos \omega_2 t] = y_2(t) \Rightarrow \therefore \dot{x}_2 = y_2(t)$$

$$\frac{d}{dt} y_2(t) = \omega_2 [-a_2 \omega_2 \cos \omega_2 t - b_2 \sin \omega_2 t] = -\omega_2^2 [a_2 \cos \omega_2 t + b_2 \sin \omega_2 t]$$

$$\Rightarrow \frac{d}{dt} y_2(t) = -\omega_2^2 x_2 \Rightarrow \text{Verified for } S_2$$

Part d)

d) Given $x_1(0)=2, y_1(0)=1$ for S_1 & $x_2(0)=-2, y_2(0)=0$ for S_2

$$\text{From above, } x_1(0) = a_1 \cos(0) + b_1 \sin(0) \Rightarrow x_1(0) = a_1 \Rightarrow \boxed{a_1 = 2}$$

$$\text{Also, } y_1(0) = \omega_1 (-a_1 \sin(0) + b_1 \cos(0)) = b_1 \omega_1 = 1 \Rightarrow \boxed{b_1 = \frac{1}{\omega_1}}$$

$$\text{Similarly, } x_2(0) = a_2 \cos(0) + b_2 \sin(0) = a_2 \Rightarrow \boxed{a_2 = -2}$$

$$\text{Also, } y_2(0) = \omega_2 (-a_2 \sin(0) + b_2 \cos(0)) = b_2 \omega_2 \Rightarrow b_2 \omega_2 = 0 \Rightarrow \boxed{b_2 = 0}$$

Part e)

Define Parameters and Solutions

```
In[ ]:= ClearAll["Global`*"];

ω1 = 2;
ω2 = 3;

(* constants from part (d) *)
a1 = 2;  b1 = 1;
a2 = -2; b2 = 0;

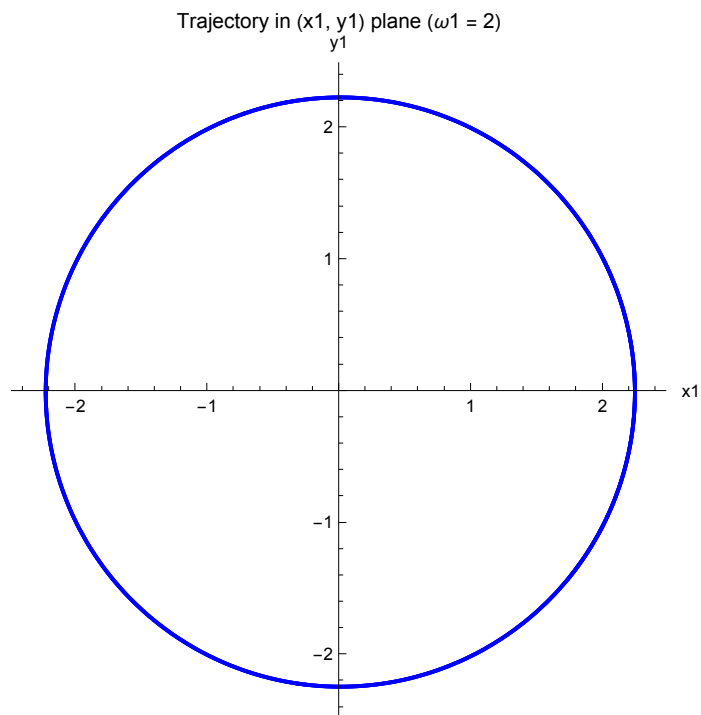
(* solutions *)
x1[t_] := a1 Cos[ω1 t] + b1 Sin[ω1 t];
y1[t_] := -a1 Sin[ω1 t] + b1 Cos[ω1 t];

x2[t_] := a2 Cos[ω2 t] + b2 Sin[ω2 t];
y2[t_] := -a2 Sin[ω2 t] + b2 Cos[ω2 t];
```

For System S1

```
In[ ]:= ParametricPlot[  
  {x1[t], y1[t]},  
  {t, 0, 10},  
  AxesLabel → {"x1", "y1"},  
  PlotLabel → "Trajectory in (x1, y1) plane ( $\omega_1 = 2$ )",  
  AspectRatio → 1,  
  PlotStyle → {Blue, Thick},  
  PlotRange → All  
]
```

Out[]:=

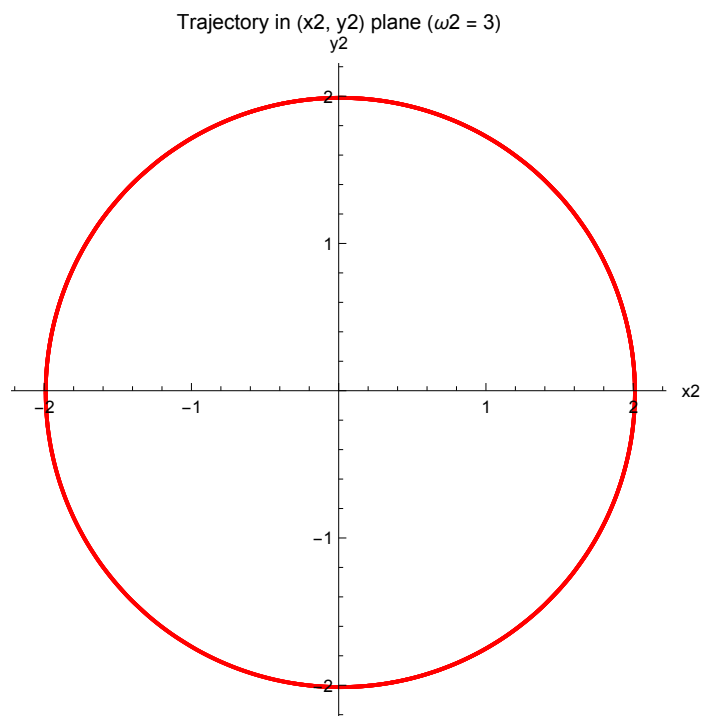


```

In[ ]:= ParametricPlot[
  {x2[t], y2[t]},
  {t, 0, 10},
  AxesLabel → {"x2", "y2"},
  PlotLabel → "Trajectory in (x2, y2) plane ( $\omega_2 = 3$ )",
  AspectRatio → 1,
  PlotStyle → {Red, Thick},
  PlotRange → All
]

```

Out[]:=



This confirms the presence of Centres

Question 5

Part a)

Note: Typo
below. Fix: y_1
(0) =
 $(\omega_1)^{-1}$

Q5) From Q4), $x_1(0)=2$, $y_1(0)$, $x_2(0)=-2$ & $y_2(0)=0$

The eq^{ns} are $x_1(t)=2\cos\omega_1 t - \sin\omega_1 t$, $x_2(t)=-2\cos\omega_2 t$
 $\therefore$ the slice-trajectory in the (x_1, x_2) plane is : $t \mapsto (x_1(t), x_2(t)) = (2\cos\omega_1 t - \sin\omega_1 t, -2\cos\omega_2 t)$

a) i) Given, $\omega_1=2$, $\omega_2=5 \Rightarrow \omega_1/\omega_2$ is rational.

ii) Given, $\omega_1=1$, $\omega_2=\sqrt{2} \Rightarrow \frac{\omega_1}{\omega_2}$ is irrational

From the plots, we see that the trajectory forms 'a closed loop', while in the 2nd case, we see that the system never returns exactly to its initial state $\Rightarrow$ No loop is formed. This is in line with what we expect:

As in (i) $\omega_1/\omega_2 \in \mathbb{Q}$, $\exists$ some common time period 'T' where both oscillators complete n_1 & n_2 cycles respectively, where $n_1, n_2 \in \mathbb{Z}$. When this ratio is irrational, the trajectories never 'close onto each other' and will create new 'arcs' in the phase space $\Rightarrow$ it "fills" the accessible region densely $\Rightarrow$ Trajectory is quasiperiodic.

```

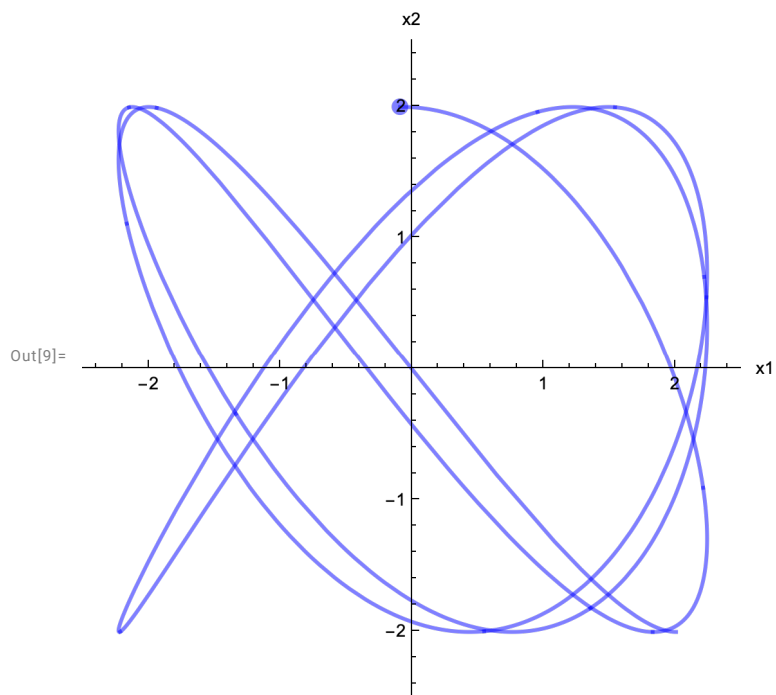
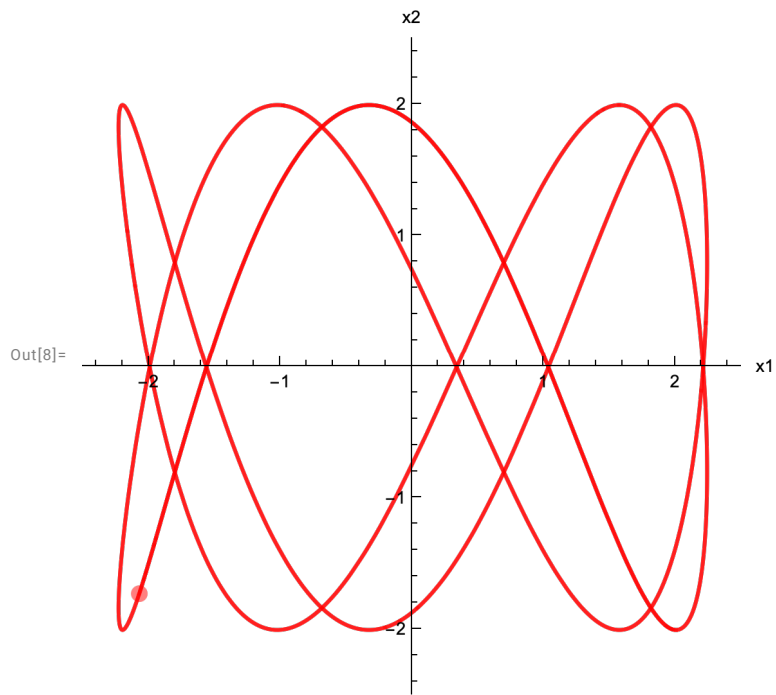
In[1]:= (* Initial Conditions from Q4*)
x10 = 2;
x20 = -2;
y20 = 0;
y10 = 1;

x1[t_, w1_] := x10 Cos[w1 t] - y10 Sin[w1 t];
x2[t_, w2_] := x20 Cos[w2 t] - y20 Sin[w2 t];

(*LLM used to refine function: to add the dot at the end of the trajectory*)
traj[w1_, w2_, tmax_, color_] := {
  ParametricPlot[
    {x1[t, w1], x2[t, w2]},
    {t, 0, tmax},
    PlotRange → {{-2.5, 2.5}, {-2.5, 2.5}},
    AxesLabel → {"x1", "x2"},
    PlotStyle → Directive[color, Opacity[0.5]]
  ],
  Graphics[
    {
      Directive[color, Opacity[0.5]],
      PointSize[0.025],
      Point[{x1[tmax, w1], x2[tmax, w2]}]
    }
  ]
};

Show[traj[2, 5, 20, Red]]
Show[traj[1, Sqrt[2], 20, Blue]]

```



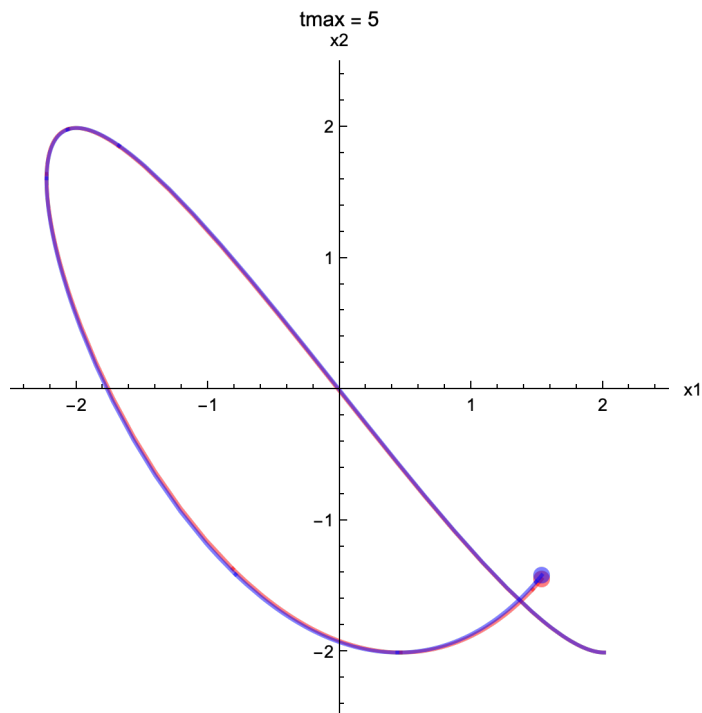
Part b)

For $t=5$, the trajectories of the rational ratio and the irrational ratio are very similar, and they don't diverge given the small time.

In[10]:=

```
Show[
  traj[1, 1.41, 5, Red],
  traj[1, Sqrt[2], 5, Blue],
  PlotLabel -> "tmax = 5"
]
```

Out[10]=

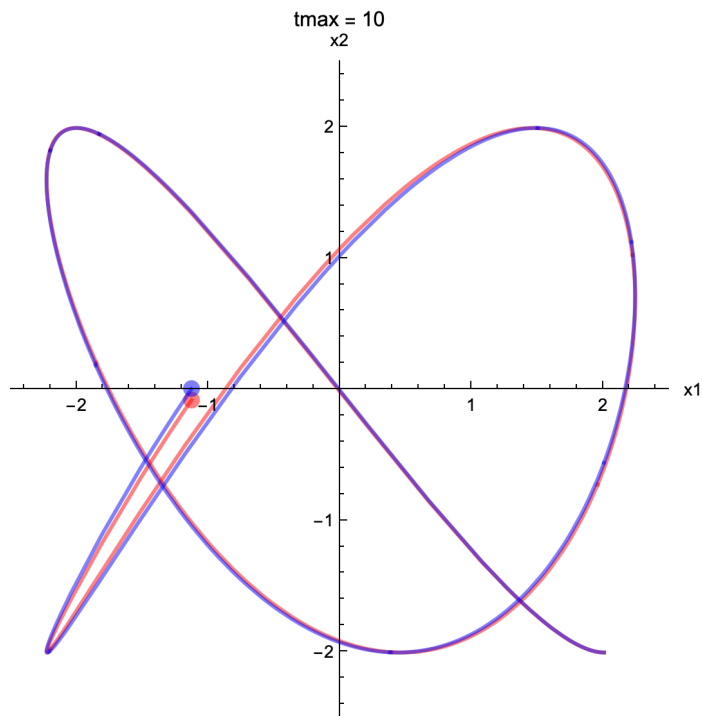


For $t=10$, the trajectories of the rational ratio and the irrational ratio are still similar, but they diverge more given the larger value of t .

In[11]:=

```
Show[
  traj[1, 1.41, 10, Red],
  traj[1, Sqrt[2], 10, Blue],
  PlotLabel -> "tmax = 10"
]
```

Out[11]=

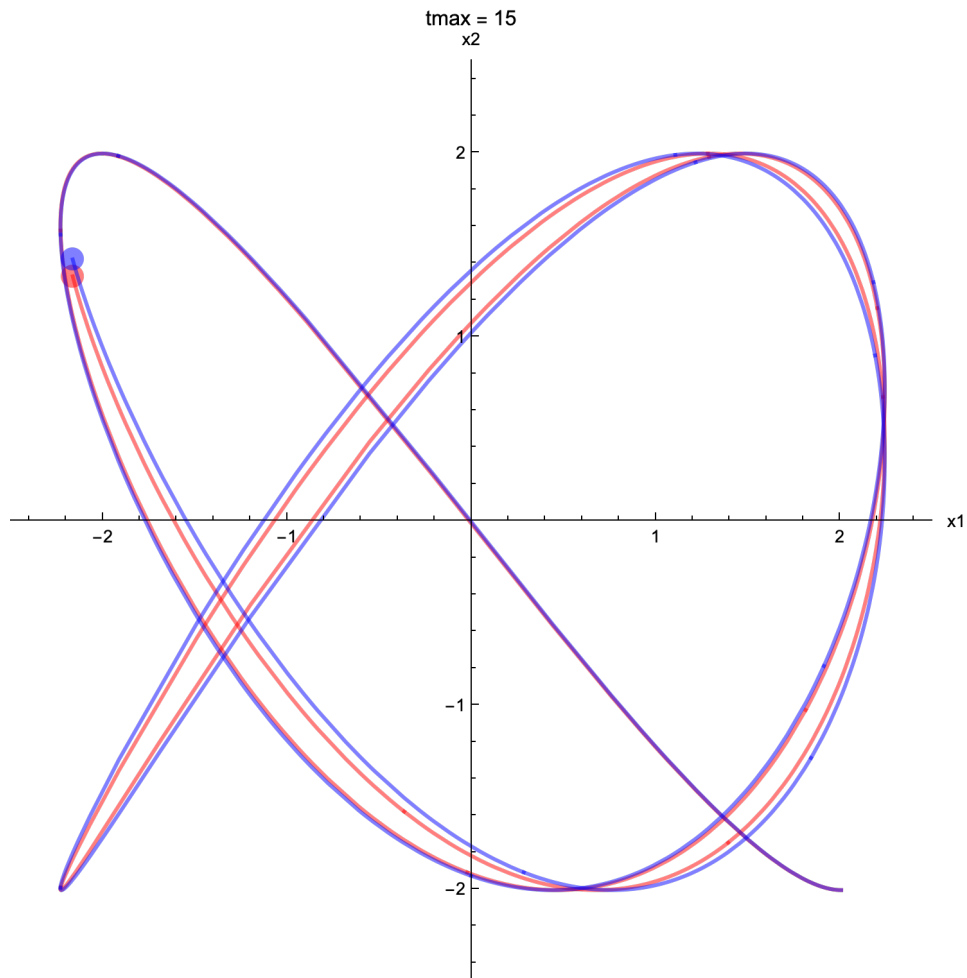


The trend of larger divergence with a greater value of t continues with $t = 15$

In[12]:=

```
Show[
  traj[1, 1.41, 15, Red],
  traj[1, Sqrt[2], 15, Blue],
  PlotLabel -> "tmax = 15"
]
```

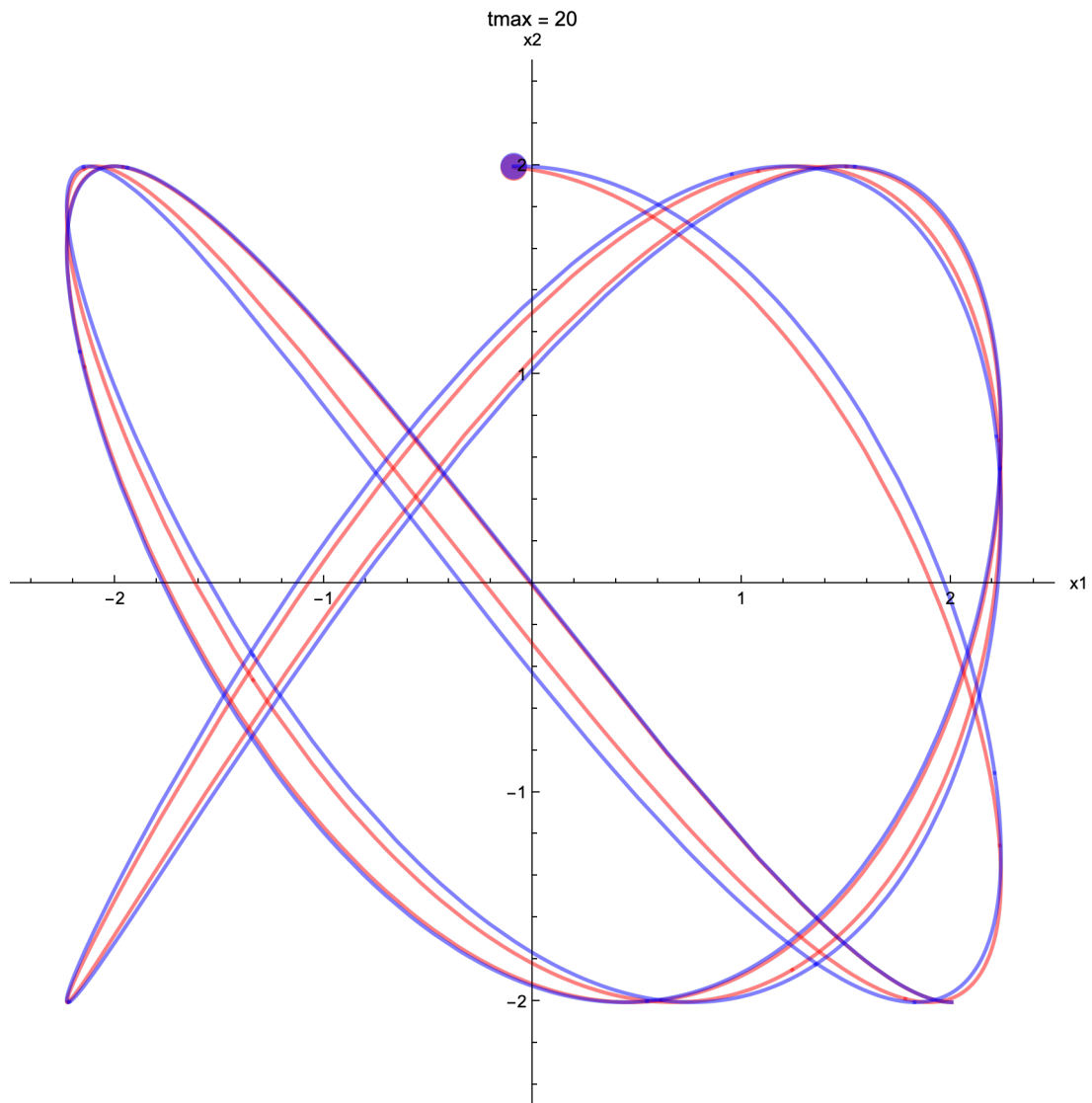
Out[12]=



For $t = 20$, we see the same, with the divergence greater

```
In[13]:= Show[
  traj[1, 1.41, 20, Red],
  traj[1, Sqrt[2], 20, Blue],
  PlotLabel -> "tmax = 20"
]
```

Out[13]=



Part c)

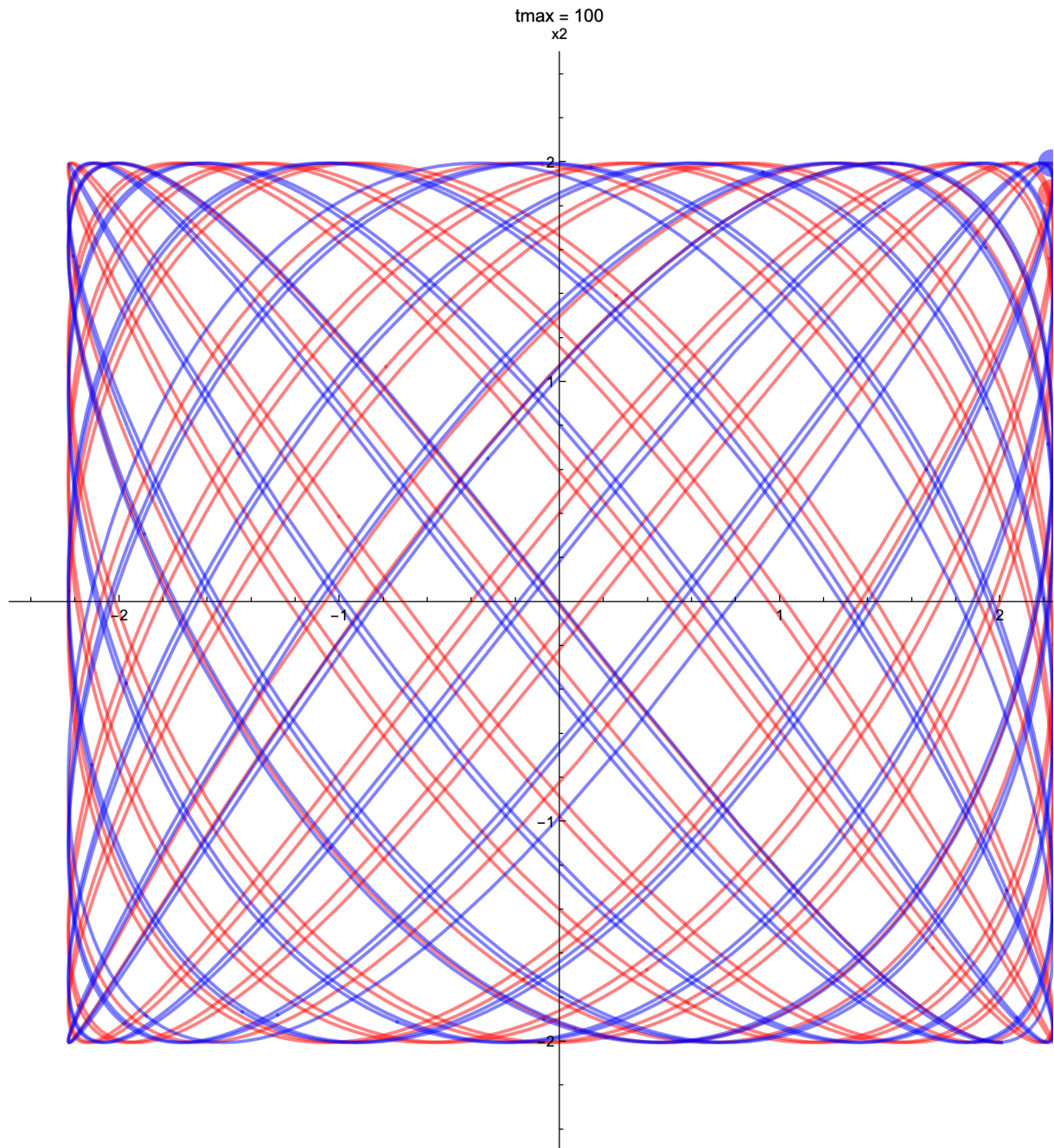
For $t = 100$, we see that the divergence is much greater here. We can also see that the rational fraction's trajectory forms a periodic motion, as expected by the analysis conducted earlier. As also expected from the analysis, we see that the irrational ratio's trajectory forms a quasiperiodic motion.

```

In[14]:= Show[
  traj[1, 1.41, 100, Red],
  traj[1, Sqrt[2], 100, Blue],
  PlotLabel -> "tmax = 100"
]

```

Out[14]=



The Quasiperiodic nature is visible with a greater value of t . Note that the rational trajectory continues to retrace its closed path.

```
In[17]:= Show[  
  traj[1, Sqrt[2], 200, Blue],  
  traj[1, 1.41, 200, Red],  
  PlotLabel -> "tmax = 200"  
]
```

Out[17]=

